

Stablecoins

Evaluating The Future of Internet Payment Protocols



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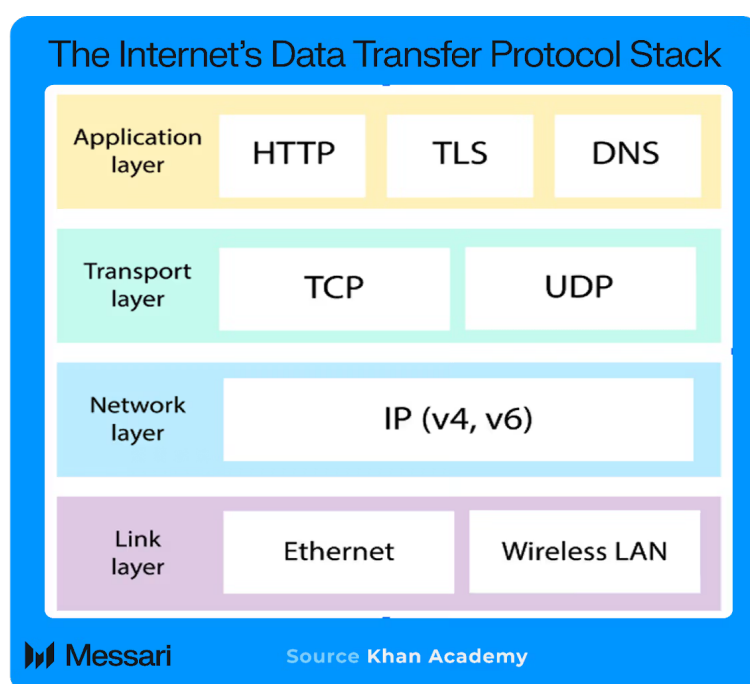
Key Insights:

- **The original internet protocols (e.g., HTTP, TCP/IP) enable data exchange but lack built-in mechanisms for sending and settling value.** As a result, payments rely on centralized intermediaries that introduce cost, latency, and compliance friction.
- **Current systems rely on human input at most steps, limiting automation opportunities.** Existing internet protocols must be improved so autonomous agents can handle payments end-to-end.
- **Stablecoins provide the flexibility and programmability needed to support digital payments at internet scale for both humans and agents.** They are already being used for peer-to-peer payments, remittances, and commerce.

- **Internet-native payments materialize into four layers: application, coordination, routing, and settlement.**

While application, routing, and settlement layers are well-populated, coordination remains underdeveloped. X402, an internet-native payment protocol, addresses this by standardizing how agents negotiate and authorize payments.

The Internet's Architectural Gap



The Internet was built for communication, not value transfer. Protocols like [TCP/IP](#), [HTTP](#), and [DNS](#) define how devices connect, how messages are formatted, and how content moves across networks. These protocols are open, meaning anyone can build on them, extend them, and rely on them without permission. This design enabled the wave of innovation we now call the web.

No such standard exists for internet payments today, leaving them fragmented and tied to individual platforms. Without a

shared protocol, developers can't build interoperable, trust-minimized systems, and users remain dependent on siloed infrastructure. To support global, programmable value exchange, the internet needs a native payment protocol. [Stablecoins](#) and emerging payment frameworks are now making this possible.

The Current State: SWIFT, ACH, and Card Network Protocols

Today's digital payment landscape reflects decades of retrofitting pre-internet financial infrastructure for online use, resulting in structural limitations. Traditional payment protocols demonstrate those constraints:

[Society for Worldwide Interbank Financial Telecommunication \(SWIFT\)](#) is a messaging network banks use to coordinate cross-border transfers. Swift uses the [MT protocol](#) (currently being phased out for [ISO 20022](#)) to standardize how payment instructions are communicated, but settlement depends on banking networks, local clearing houses, and time zones, among other factors. Transfers often take [1-5 business days](#) to settle, costing between [\\$15 and \\$50](#). These delays stem from the layered intermediary networks previously requisite in an era of physical document transfer, but now unnecessary with digital commerce.

[Automated Clearing House \(ACH\)](#) is a network for batch-based bank transfers in the U.S. While it supports everyday use cases like payroll and bill pay, ACH is limited by business-hour settlement windows, delayed finality (1-2 days for settlement), and jurisdictional constraints such as only processing U.S. bank payments denominated in dollars. It lacks programmability and native support for conditional payments, making it suboptimal for real-time internet

applications.

The [ISO 8583 protocol](#), used by card networks like [Visa](#) and [Mastercard](#), governs how authorization, clearing, and settlement messages are structured between issuers and acquirers. It enables rapid approval but relies on proprietary infrastructure, licensed access, and a multi-party stack that adds cost and latency. This structure limits support for microtransactions and restricts developer flexibility.

These systems work but are structurally limited by centralized control and rigid messaging formats. Additionally, they don't support peer-to-peer communication, conditional logic, or multi-agent coordination. Simply put, they were not built for the modern Internet.

The Future State: Agentic Payments

Jensen Huang, CEO of NVIDIA predicts agents will be a “multi-trillion dollar opportunity”

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As the internet moves toward machine-mediated interaction,

autonomous agents are positioned to become a major interface for initiating and coordinating payments. Agentic payments represent a natural progression in the evolution of internet architecture. Websites gave rise to APIs (a set of rules that allow software to communicate and share functionality), which enabled companies to automate data retrieval. Agents take this a step further with superior execution and abstraction, allowing software to transact directly on behalf of users or systems.

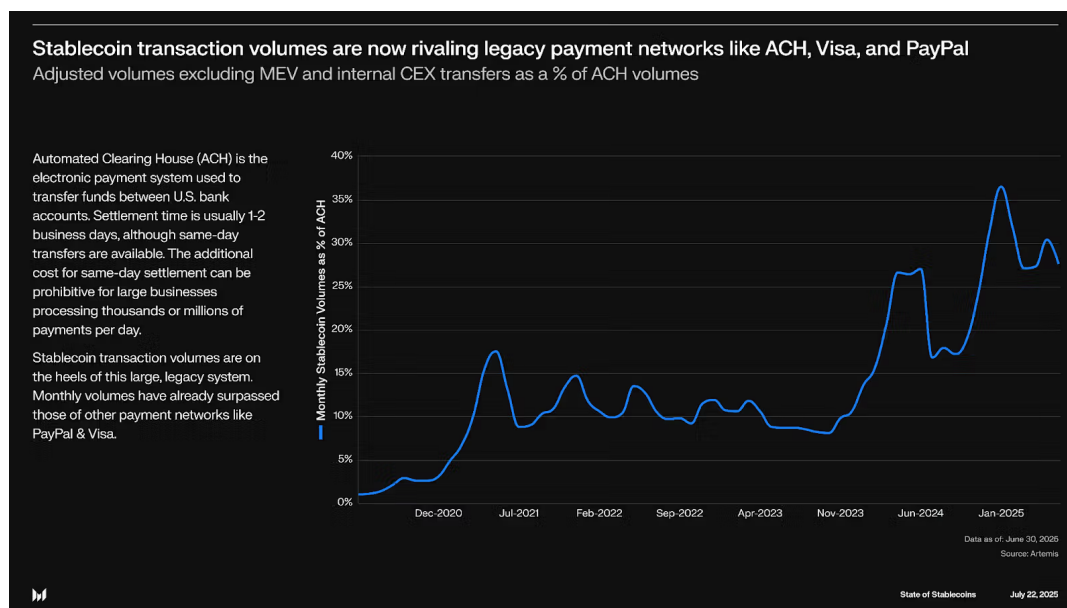
Today, legacy payment systems lack the standardized protocols (e.g., how [SMTP](#) standardized email coordination) that allow agents to make payments across the internet without custom integrations. Today's payment systems rely on proprietary technology that is not machine-readable. Without shared standard protocols, autonomous agents have no reliable way to coordinate payments, making agentic commerce inoperable in the current environment.

While some agents can interact with frontends (e.g., [OpenAI Operator](#)), doing so introduces latency and higher computing costs, as these systems are built for manual use rather than automated execution. Humans need UI-based infrastructure to navigate workflows, but this model is poorly suited for autonomous agents. Agents operate best in machine-readable environments, allowing them to handle workflows without relying on UI logic.

The infrastructure for agentic payments is already emerging. Stablecoins are the ideal asset for utilizing this technology in an internet-native fashion.

Stablecoins: The Substrate for Native Payments

Internet-native payment protocols require a digital asset with the stability of fiat and the benefits of blockchain. Stablecoins meet this need by linking legacy finance with blockchain infrastructure.



In less than five years, [stablecoin volume](#), adjusted for MEV and internal CEX transfers, reached 35% of ACH volumes. Total stablecoin volume was \$18.4 trillion at the close of 2024, a 140% increase YoY, while ACH volume was \$85.6 trillion, a 7% increase YoY. Stablecoin adoption is growing as major institutions like Visa, Mastercard, and PayPal embrace the technology for faster, cheaper payments.

Advantages Over Legacy Payment Infrastructure

Stablecoins have four advantages that make them uniquely suited to become internet payment primitives:

Microtransaction Viability: Stablecoins support payments as

small as fractions of a cent, unlike traditional systems with high fixed fees that make microtransactions (e.g., \$0.30+ minimums on [Stripe](#) or [PayPal](#)) impractical. This unlocks new business models for the internet, including pay-per-use access to content, data, APIs, or compute.

Transparent and Open Reputation Systems: Stablecoin transactions occur on public ledgers, allowing open auditability without relying on private credit bureaus or opaque scoring systems.

Global, Permissionless Access: Anyone with internet access and a wallet can use stablecoins, bypassing geographic, banking, and credit barriers. This open access could be vital for [unbanked and underbanked populations](#), who are often excluded from traditional financial systems due to a lack of documentation, low credit scores, or high account maintenance costs.

Composability and Programmability: Stablecoins easily integrate with other smart contract protocols, enabling modular financial systems where applications compose like APIs across software stacks. This composability allows one protocol's output to become another's input, which creates a capital-efficient chain of applications that work together permissionlessly.

Stablecoin infrastructure improves on legacy internet payments by enabling faster settlement and blockchain-based accounting. However, the design choices of major issuers like [Circle](#) and [PayPal](#) limit their ability to operate as true internet-native primitives. These issuers retain vertical control over issuance, redemption, and access through centralized APIs, meaning that while settlement may occur on public blockchains, the surrounding infrastructure remains closed. A

potential alternative would be to design a modular, internet-native payments stack.

The Internet-Native Payments Stack

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Routing Layer

Finds a valid path for value delivery and ensures the message (and value) gets to the right place.

D

Settlement Layer

The settlement layer is where value transfer is finalized, recorded, and made irreversible using cryptographic proof and consensus-based finality.

C

Routing Layer

D

Settlement Layer

Interledger Protocol

LayerZero

Wormhole

Ethereum

Solana

Stable Plasma

Data as of: July 31, 2025

Source: Messari

Messari

A truly internet-native payment stack separates stablecoin payments into four distinct layers, each with a specialized role. This modular architecture is inspired by the design of traditional internet protocols and enables more flexible, composable, and programmable payments across the web.

Here's what this looks like in practice:

Alice, a software agent working on behalf of a weather forecasting service, wants to pay John, who runs a decentralized API that provides high-resolution weather data for drones.

Alice queries weather data from John's service via an autonomous agent (application layer). John's weather API is gated by X402 (coordination layer), which returns an HTTP

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API

402 response requiring a small payment to access the data. The 402 response includes a machine-readable message telling Alice's agent how much to pay, what asset to use (e.g., USDC), and where to send the funds. Alice's agent is on Ethereum, while John's is on Solana, so Alice's agent must coordinate with a messaging protocol like LayerZero to bridge the funds on the condition that the requested data was delivered (routing layer). Once the routing layer ensures conditions are met, the payment is settled onchain (settlement layer). Upon receiving confirmation of payment, John's server releases the weather data to Alice's agent.

In this example, every layer plays a distinct role:

- The application layer initiates the intent.
- The coordination layer handles negotiation and authentication of intent.
- The routing layer ensures that the value can reach its destination under the right conditions.
- The settlement layer executes the actual transfer of funds.

Each layer of the internet-native payment stack exists today, but they are disconnected. The application layer is evolving to support agent-based interactions. Routing protocols, while nascent, are commonly used to bridge funds across chains. The settlement layer is populated with multiple high-throughput blockchains like Solana and Base. The coordination layer is the most underdeveloped, but it will act as the connective tissue between the application, routing, and settlement layers.

Application Layer

The application layer is the user-facing surface of the payment stack. It is where individuals, merchants, developers, and agents initiate and interact with payments. This layer includes tools like wallets, e-commerce platforms like [Shopify](#), [point-of-sale \(PoS\) systems](#), Web3 applications, agents, and service APIs. Applications are responsible for capturing intent from users and/or agents. They present UI elements like “pay,” “subscribe,” or “unlock content,” and pass this raw intent to the coordination layer.

Although the application layer does not execute payments directly, it connects with the other layers to initiate and monitor payment flows. It handles tasks like generating payment requests, collecting user signatures, and determining whether to grant access based on payment outcomes. In a modular stack, applications can flexibly integrate with different protocols across the other layers without needing to manage the underlying infrastructure themselves.

Coordination Layer

The coordination layer manages how payment instructions are structured, authorized, and resolved between participants. It sits between the application and routing/settlement layers, translating user or agent intent into standardized, verifiable instructions that downstream systems can execute. Coordination layers are not responsible for the actual movement of funds; that is the job of the settlement layer.

Rather than hardcoding flows into each application, the coordination layer provides reusable logic for payment negotiation, authorization, and validation. This layer enables payments permissionlessly, similar to internet communication protocols like HTTP or TCP/IP.

X402 and intent frameworks exemplify this design and could be key protocols in the coordination layer.

X402

[X402](#), created by Coinbase, is an open, internet-native payment protocol that enables payments using stablecoins and other crypto assets. It builds on the [HTTP status code 402](#) ("Payment Required"), which was created to enable a digital cash payment system. 402 hasn't been widely adopted because payments weren't standardized at the protocol level, meaning there is no universally shared standard that all payment systems follow to exchange money on the internet. X402 now leverages the code for websites, APIs, and agents to request payment before serving content or data. X402 is HTTP-native, meaning it runs directly on the web and is compatible with existing standards like [MCP](#), Google's [A2A](#), and even direct browser navigation to paywalled URLs.

An [API](#) (Application Programming Interface) is used to retrieve data, trigger operations, or integrate services across different systems. Interacting with APIs is often tedious, requiring

account setup, linking a payment method, and secure generation and management of API keys. X402 simplifies this by using a crypto wallet for identity and payment, letting agents or developers call and pay for APIs without manually signing up for a developer account.

X402 does not replace the need for the routing or settlement layers. Instead, it helps agents and apps coordinate using a shared standard, removing the need for custom integrations at each endpoint.

The protocol adds a single line of code to any [HTTP endpoint](#) (e.g., `example.com/weather`). When a client makes a request, the server checks for a valid payment or signature. If present, the content is served; otherwise, the server returns a structured 402 message explaining how to pay. This machine-readable format allows agents to parse, pay, and retry automatically.

For [example](#), an engineer at Base built a miniapp that allows you to post to the Base app and X at the same time. Each user gets one free X post a day; after that, posts cost 2 USDC. This conditional logic is powered by X402.

X402 removes the need for ad hoc logic at the application layer and gives agents a reliable way to negotiate access and prove payment intent. It also creates an easy path for user-facing apps and agents to connect with underlying routing and settlement systems. Ongoing development includes the "[Bazaar](#)," a discovery layer for monetized APIs that acts like a search engine for agents to dynamically find, price, and interact with services.

Routing Layer

While coordination defines what needs to happen (intent), routing determines how it can happen and delivers the message (pathfinding, ensuring a valid route exists, and asset conversion). The routing layer is similar to messaging protocols today (e.g., [LayerZero](#), [Wormhole](#), [Chainlink CCIP](#)) but should exist in tandem with the coordination layer to service agents as well.

Although the messaging protocols today provide sufficient ability to handle routing for most crypto assets, payments require a different set of qualities:

1. **Stateful Routing:** Stateful routing tracks context across a transaction, including sender, receiver, session data, and partial payment state. Messaging protocols like Wormhole and LayerZero send isolated messages without preserving this context, offering no native support for retries or fallback paths. Developers must build this logic themselves, adding complexity and the risk of errors.
2. **Conditional, Atomic Payments:** Conditional payments transfer value only when predefined conditions are met, such as receiving data or revealing a [preimage](#). Messaging bridges transmit data but not value, and they lack native support for mechanisms like timelocks or escrow. Conditionality must be implemented in separate smart contracts, introducing reliance on offchain messaging and external trust. Without additional coordination layers, atomicity and payment guarantees remain fragile.
3. **Ledger-Agnostic Interoperability:** This refers to the ability to move value between different types of systems, like from a blockchain wallet to a bank account.

Messaging protocols like Wormhole and LayerZero are chain-centric and only work across compatible smart contract platforms. They cannot route payments to banks, custodians, or other non-chain systems without custom integrations, limiting their flexibility and global reach.

Modern cross-chain messaging protocols like Wormhole, LayerZero, and Axelar are optimized for delivering messages across chains, but they lack the qualities above. Messaging layers push this burden up to the application layer, where developers must build their own logic on a per-use-case basis.

This fragmentation resembles the early internet before standards like TCP/IP for HTTP. Web developers couldn't connect systems without shared transport or communication standards. Today, app developers face similar challenges routing payments without a standardized coordination layer. Humans can and have worked around this, but agents cannot operate reliably on inconsistent systems.

A potential solution to this gap is the [Interledger Protocol \(ILP\)](#), a payment-specific routing protocol inspired by internet communication protocols. Like how [IP routes data across networks](#), ILP breaks payments into standardized [packets](#), which are small, uniformly formatted payment instructions that can be processed independently. These packets are forwarded through a series of intermediaries called connectors that facilitate payment forwarding between different ledgers or networks.

Each connector escrows funds temporarily and only releases them when a cryptographic condition is fulfilled. This design enables stateful routing, where each step maintains awareness of the transaction's progress, and conditional payments, where

value only moves if all specified criteria are met.

Importantly, ILP is ledger-agnostic, working across blockchains (e.g., [Ethereum](#), [Solana](#)), banks, and custodial systems. Its ability to abstract settlement differences and enforce delivery guarantees makes it especially compelling for agentic commerce, where autonomous systems require deterministic, cross-network payment infrastructure.

The routing layer exists in fragmented form today, through messaging protocols and ledger-agnostic systems like ILP. However, for agents to use it effectively, it must be connected to both the application and settlement layers through a coordination layer that defines payment structure, conditions, and access logic.

Settlement Layer

The settlement layer is where payments reach finality. The underlying infrastructure irreversibly records transactions and enforces the transfer of funds. Once a payment is settled here, it is considered complete and tamper-proof.

This layer includes public blockchains such as [Ethereum](#) and [Solana](#), specialized systems like stablecoin platforms (e.g., USDC on Base or Solana), and offchain rollups or stablechains like [Stable](#) and [Plasma](#). Each provides a different tradeoff between speed, cost, and security guarantees, but all serve the role of final settlement.

Coordination and routing layers rely on the settlement layer to commit payment outcomes and verify delivery. Whether an agent uses X402 to negotiate access or ILP to route across networks, actual value transfer is finalized at this layer.

Settlement anchors programmable, internet-native payments

to a shared state of truth.

Impact of Moving to Internet-Native Payments

Internet-native payment protocols represent a structural shift in how value moves online. Today's systems rely on platform-specific integrations, proprietary APIs, and intermediaries that impose fees, delays, and geographic restrictions. Moving to open, programmable payment protocols replaces these limitations with a universal, permissionless infrastructure for value transfer.

This shift mirrors the internet's past transitions. As protocols like HTTP and TCP/IP replaced siloed communication systems, payment protocols make economic interaction interoperable and composable. Developers gain the ability to embed payment logic directly into applications, workflows, or smart contracts, without relying on centralized services. This includes conditional payments, automated settlements, and native support for multi-party coordination.

Importantly, internet-native payments also lay the groundwork for agentic systems. Agents cannot rely on frontends or navigate fragmented APIs. They require deterministic coordination and verifiable messaging standards, which internet-native protocols enable. While human users benefit from lower costs and more flexible tooling, agents depend on this infrastructure to operate reliably. Ultimately, internet-native payments will bring permissionless value transfer built on stablecoins to a global scale.

This report was co-authored by Lincoln Murr, a member of

Coinbase Developer Platform, who contributed to X402.

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